[INCH-POUND]

H-B-651J
June 11, 1990
TO SUPERSEDE
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April 14, 1982

FEDERAL SPECIFICATION

BRUSHES, FLOOR SWEEPING

This Federal Specification is approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers floor sweeping brushes, without handles, with natural and synthetic filler set in wood or plastic blocks.

1.2 Classification. Floor sweeping brushes covered by this specification shall be of the following classes and sizes as specified (see 6.2).

1.2.1 Classes.

Class 1 - Fine natural fiber or fine polyester filament.

Class 2 - Tampico fiber or medium polypropylene filaments, and either fine natural fiber or fine polyester filament. (combined old class 2 and class 3).

Class 4 - Medium polypropylene filaments.

Class 5 - Heavy duty polypropylene filaments.

1.2.2 Sizes. Block length.

Size 14 - 14 inches
Size 18 - 18 inches
Size 24 - 24 inches
Size 30 - 30 inches
Size 36 - 36 inches

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this documentshould be sent to: General Services Administration, Federal Supply Service, General Products Commodity Center, General Engineering Group (7FXEG), 819 Taylor St., Fort Worth, TX 76102

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2. APPLICABLE DOCUMENTS

2.1. Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues used shall be those in effect on the date of the solicitation or request for bids.

Federal Specifications:

NN-H-104 - Handles, Wood, ACME Threaded and Tapered End
PPP-B-1880 - Brushes, Preparation for Delivery of

Federal Standard

FED-STD-376 - Preferred Metric Units for Use by the Federal Government

Single copies of this document and other Federal Specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection

Copies of Military Specifications and Standards are available from Military Specifications and Standards, Building 4, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Wood blocks. Wood blocks shall be fine textured, close grained and straight grained hardwood.

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3.1.2 Plastic blocks. Plastic for molding blocks shall be general purpose impact resistant plastic molding material commercially used for floor brushes.

3.1.3 Drawing wire. Drawing wire shall be not less than 0.015 inch diameter brass wire.

3.1.4 Staple wire. Staple wire shall be not less than 0.034 inch diameter steel staple wire.

3.1.5 Filler material. The brush filler shall be as specified. The weight of the filler material in each brush shall be as specified in table I. The weight specified shall be of the trimmed filler as removed from the end item.

3.1.5.1 Fine natural fiber. Fine natural fiber shall be either 100 percent horsetail hair as specified in 3.1.5.1.1 or a 50/50 mixture of horsetail hair as specified in 3.1.5.1.1 and a vegetable fiber such as cocoa fiber, at the manufacturers option.

3.1.5.1.1 Horsetail hair. Horsetail hair shall be extra stiff, natural horsetail hair, cleaned and sterilized using boiling water, dry heat, or autoclaving.

3.1.5.2 Synthetic filament. The synthetic filament shall be polyester or polypropylene material. The color of the filament shall be at the manufacturers option.

3.1.5.2.1 Fine polyester filament. Shall be a level or crimped polyester filament meeting the following requirements:

Cross-section - Round
Diameter (average) - 0.014 inch, +/- 0.0005 inch

3.1.5.2.2 Medium polypropylene filament. Shall be a level or crimped polypropylene filament meeting the following requirements:

Cross-section - Round
Diameter (average) - 0.014 inch, +/- 0.0005 inch

3.1.5.2.3 Heavy-duty polypropylene filament. Shall be a level or crimped polypropylene filament of two diameters, mixed, either diameter shall not be less than 45 percent of the total by count with the remainder being the other diameter. The filaments shall meet the following requirements:

Cross-section (both) - Heavy X
Diameters (average) - 0.018 inch, +/- 0.0005 inch, for the lesser diameter filament
0.028 inch, +/- 0.0005 inch, for the greater diameter filament

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3.1.5.3 Tampico fiber. All tampico fiber shall be of natural or black dyed color.

3.2 Construction. The construction of the brushes shall conform to the requirements specified in table I. The brushes shall be constructed with either staple set or wire drawn tufts.

Table I, Physical Requirements.

	Brush Size				
	14	18	24	30	36
Block					
Length (inches, minimum)	13-3/4	17-3/4	23-3/4	29-3/4	35-3/4
Width at brush face (inches, minimum)	--------- 2-7/16 for all sizes -------				
Thickness					
Wood (inches, minimum)	---------- 15/16 for all sizes -------				
Plastic (inches, minimum)	---------- 7/8 for all sizes --------				
Depth of tuft holes					
Class 1-4 (inches, min.)	---------- 5/16 for all sizes --------				
Class 5 (inches, minimum)	---------- 7/16 for all sizes --------				
Total number of holes, minimum, all classes	162	203	228	346	416
Filament length extending from block, all classes (inches, minimum)	---------- 3 for sll sizes ---------				
Filament weight per brush, (ounces, minimum)					
Class 1-4	7-19/32	9-3/4		16-9/32	19
Class 5		10-9/16	13-13/16		

3.2.1 Block. The block shall be fabricated of either wood as specified in 3.1.1 or plastic as specified in 3.1.2. The block shall have a rounded or flat top. The block shall be one or two piece construction as required. The two piece block shall be used only for wire drawn brushes. The two piece block shall be fastened together with rivets or bolts using lock washers and nuts or washers and locknuts. Two ACME threaded holes shall be provided for handles in the top of each block, centered to within 1/16 inch of the longitudinal center. The holes shall be drilled through the block, or to a minimum depth of 1-1/16 inch, each to be at an angle of 45 degrees, +/- 10 degrees, from the vertical, and shall be threaded with an ACME thread having 5 threads per inch. The holes and threads shall be of such dimensions so as to enable a handle with threads as specified in NN-H-104, figure 1, to be attached (see 4.2.2.2).

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3.2.2 Filler.

3.2.2.1 Class 1. The class 1 brushes shall be made from fine natural fiber (see 3.1.5.1) or fine polyester filament (see 3.1.5.2.1) at the option of the contractor.

3.2.2.2 Class 2. The class 2 brushes shall be made from a mixture of 45 to 55 percent fine natural fiber (see 3.1.5.1) or fine polyester filament (see 3.1.5.2.1), with the remainder being tampico fiber (see 3.1.5.3) or medium polypropylene filament (see 3.1.5.2.2).

3.2.2.4 Class 4. The class 4 brushes shall be made from medium polypropylene filament (see 3.1.5.2.2).

3.2.2.5 Class 5. The class 5 brushes shall be made from a heavy duty polypropylene filament (see 3.1.5.2.3).

3.2.3 Set. The brushes shall be either staple set or wire drawn. The tufts shall not loosen, break, or pull out of the brush when tested in accordance with paragraph 4.2.2.1.

3.2.3.1 Stapled. The filler material shall be cut to double length, folded in the center, and stapled. The staples shall be made of the wire specified in 3.1.4.

3.2.3.2 Wire drawn. Continuous lengths of drawing wire as specified in 3.1.3 shall be used. The drawing wire in the finished brush shall be concealed within the block.

3.3 Trim. The finished brush filler material shall be evenly and uniformly trimmed to conform to the dimensional requirements of the finished brushes.

3.4 Wood finish. All wood block brushes shall have the wood components covered with at least one coat of clear protective finish.

3.5 Marking. Each block shall be clearly and permanently marked with its National Stock Number (NSN) and the manufacturers name or recognized trademark.

3.6 Workmanship. The brushes shall be free from defects which may affect their serviceability, durability or appearance. The brushes shall be neatly constructed and finished and shall have no sharp edges. The filler material shall be free from foreign matter and loose material.

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3.7 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing employees to undue hazards from the recovered materials. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.8 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified in this specification. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections or tests set forth in the specifications where such action is deemed necessary to assure that supplies and services conform to the prescribed requirements. The contractor is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Inspection and tests. Samples for inspection and tests shall be selected in accordance with MIL-STD-105. A lot shall be all brushes of one class and size submitted for inspection at one time. A sample unit shall consist of one brush.

4.2.1 Inspection of end item. Examination of the end item shall be made for defects listed in table II and for physical and dimensional characteristics listed in table I. The inspection level shall be level I with an Acceptable Quality Level (AQL) of 4.0 defects per hundred units.

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Table II. Visual examination.

Examine	Defect
Brush	Not class or size specified.
Finish	Any rough or sharp edges, burrs, runs, foreign matter, chips, scratches, blisters, flaking or peeling.
	Wood not coated as specified.
Block	More than two pieces.
	More than one piece for staple set filler.
	Two piece block not firmly fastened together as specified (see 3.2.1).
	Does not have a rounded or flat top.
	Does not have two holes as specified (see 3.2.1).
	Threaded holes have threads damaged or missing.
	Not marked as specified in 3.5.
Workmanship	Loose grain or rough edges on wood block.
	Block not smoothly finished.
	Brush or block not clean.
Marking	Missing, illegible or incorrect.

4.2.1.1 Dimensional examination. Width shall be measured on the face of the block at not less than three places. Thickness shall be measured at not less than three places, two of which shall be measured within 1 inch of each end. Failure at any point fails the sample.

4.2.2 Tests. The inspection level for testing shall be S-2. Each sample unit shall be tested in accordance with the procedures outlines herein.

4.2.2.1 Set retention. Grasp a complete tuft with a suitable holding device. With the block secured apply a force of 4 pounds, by use of a spring scale or weight, in the direction perpendicular to the face of the block. The tuft shall not break, loosen or pull free. Any failure shall fail the lot.

4.2.2.2 ACME threaded hole. A handle with threads conforming to NN-H-104, figure 1, shall be screwed into both holes of each sample handle. Any inability to be inserted or looseness when fully screwed in shall fail the lot.

4.3 Examination for preparation for delivery. A random sample of unit, intermediate and shipping containers (as applicable) shall be selected from each lot and examined for conformance with the preservation, packaging, packing, labeling and marking required in the contract or order. Samples shall be selected in accordance with MIL-STD-105, Inspection level S-2, AQL 6.5 expressed in terms of defects per hundred unit.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, labeling, and marking. In addition to the marking required by paragraph 3.5, the preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

5.2 Natural fibers. Brushes containing any amount of natural fibers shall be packed with a suitable amount of napthalene flakes or balls.

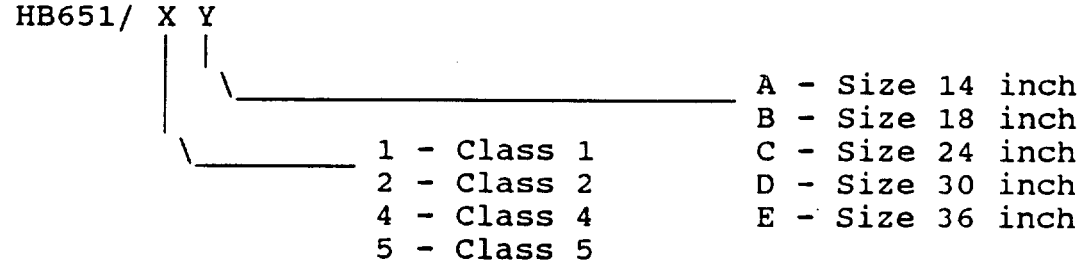
6. NOTES

6.1 Intended use. Classes 1 through 4 floor sweeping brushes are light duty brushes intended for use in sweeping wood, smooth concrete, tile and other types of smooth flooring inside buildings. Class 5 floor sweeping brushes are heavy duty brushes intended for sweeping concrete garage floors, basements and sidewalks.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following on procurement documents:

(a) Title, number, and date of this specification.
(b) Class and size of brush required (see 1.2).
(c) Packaging and packing (see section 5).
(d) Marking required (see 5.3).

6.3 Part numbering system. The following part numbering system shall apply to this specification:



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6.4 Referencing NSN's. The following NSN's reference this document (this may not be a complete list):

NSN	Class and Size	Part No.
7920-00-243-3407	Class 1, Size 18 inch	HB651/1B
7920-00-292-2363	Class 1, Size 30 inch	HB651/1D
7920-00-292-2365	Class 2, Size 14 inch	HB651/2A
7920-00-292-2367	Class 2, Size 18 inch	HB651/1B
7920-00-264-4638	Class 2, Size 30 inch	HB651/2D
7920-00-292-2362	Class 2, Size 36 inch	HB651/2E
7920-00-965-4887	Class 3, Size 18 inch	HB651/3B
7920-00-965-4886	Class 4, Size 18 inch	HB651/4B
7920-00-292-2359	Class 5, Size 18 inch	HB651/5B
7920-00-292-2360	Class 5, Size 24 inch	HB651/5C

6.5 Handles. These brushes do not include handles. The handle intended to be used with these brushes is NSN 7920-00-263-0328.

6.6 Supersession data. What were formerly classes 2 and 3 have been combined into class 2. Each class will not have all sizes available.

MILITARY INTERESTS:

NONE: DoD has determined that no military activity has an official interest in this Federal Specification.

Civil Agency Coordinating Activity:
JUS-FPI

Preparing Activity:
GSA-FSS

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